

CHAIN-OF-THOUGHT PROMPTING ELICITS REASONING IN LARGE LANGUAGE MODELS

Google Research, Brain Team (2022, Oct)

딥러닝 논문읽기 자연어처리팀

팀원 : 김지연, 김재민, 이기성,
주정헌(발표자)

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Introduction

- **Revolution in NLP** : Scaling up the size of language models
(aka. GPT3, PaLM, Lamda)
- **Challenge** : Not sufficient in “*arithmetic, commonsense, and symbolic*” reasoning

CHAIN OF THOUGHT PROMPTING

1. Decompose multi-step problems into intermediate steps.
- 2 : Providing secondary window of model before answering final question.
3. math, symbolic, and commonsense reasoning are applicable to solve via language.
4. Eliciting(이끌어내다) right answer by few-shot prompting

Standard prompting

Input: Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: The answer is 11.

...

Q: John takes care of 10 dogs. Each dog takes .5 hours a day to walk and take care of their business. How many hours a week does he spend taking care of dogs?
A:

Model output: The answer is 50. ❌

Chain of thought prompting

Input: Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

...

Q: John takes care of 10 dogs. Each dog takes .5 hours a day to walk and take care of their business. How many hours a week does he spend taking care of dogs?
A:

Model output: John takes care of 10 dogs. Each dog takes .5 hours a day to walk and take care of their business. So that is $10 \times .5 = 5$ hours a day. 5 hours a day $\times$ 7 days a week = 35 hours a week. The answer is 35 hours a week. ✅

Arithmetic Reasoning

Experiments

Benchmarks : 5 math word problems.

Standard prompting(baseline): GPT3

CoT prompting : Multiple cases problems

Math Word Problems (free response)

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Math Word Problems (multiple choice)

Q: How many keystrokes are needed to type the numbers from 1 to 500?
Answer Choices: (a) 1156 (b) 1392 (c) 1480 (d) 1562 (e) 1788

A: There are 9 one-digit numbers from 1 to 9. There are 90 two-digit numbers from 10 to 99. There are 401 three-digit numbers from 100 to 500. $9 + 90(2) + 401(3) = 1392$. The answer is (b).

CSQA (commonsense)

Q: Sammy wanted to go to where the people were. Where might he go?
Options: (a) race track (b) populated areas (c) desert (d) apartment (e) roadblock

A: The answer must be a place with a lot of people. Race tracks, desert, apartments, and roadblocks don't have a lot of people, but populated areas do. So the answer is (b).

StrategyQA

Q: Yes or no: Would a pear sink in water?

A: The density of a pear is about 0.6 g/cm^3 , which is less than water. Thus, a pear would float. So the answer is no.

Date Understanding

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.

Sports Understanding

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.

SayCan (Instructing a robot)

Human: How would you bring me something that isn't a fruit?

Explanation: the user wants something to eat that isn't a fruit. An energy bar is not a fruit, so I will bring the user an energy bar.

Plan: 1. find(energy bar) 2. pick(energy bar) 3. find(user) 4. put(energy bar) 5. done().

Last Letter Concatenation

Q: Take the last letters of the words in "Lady Gaga" and concatenate them.

A: The last letter of "Lady" is "y". The last letter of "Gaga" is "a". Concatenating them is "ny". So the answer is ny.

Coin Flip (state tracking)

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

Result

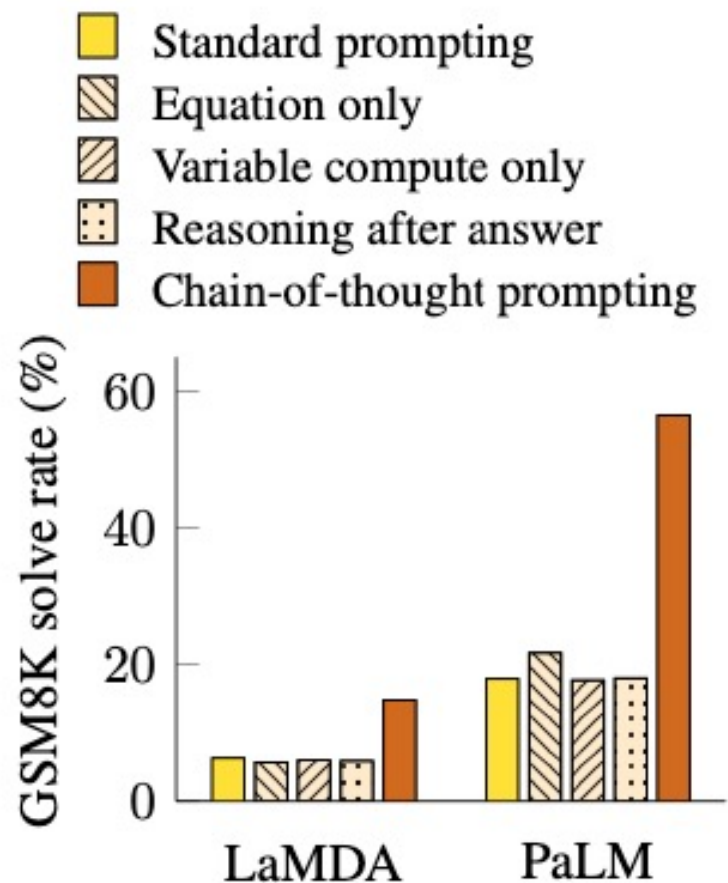
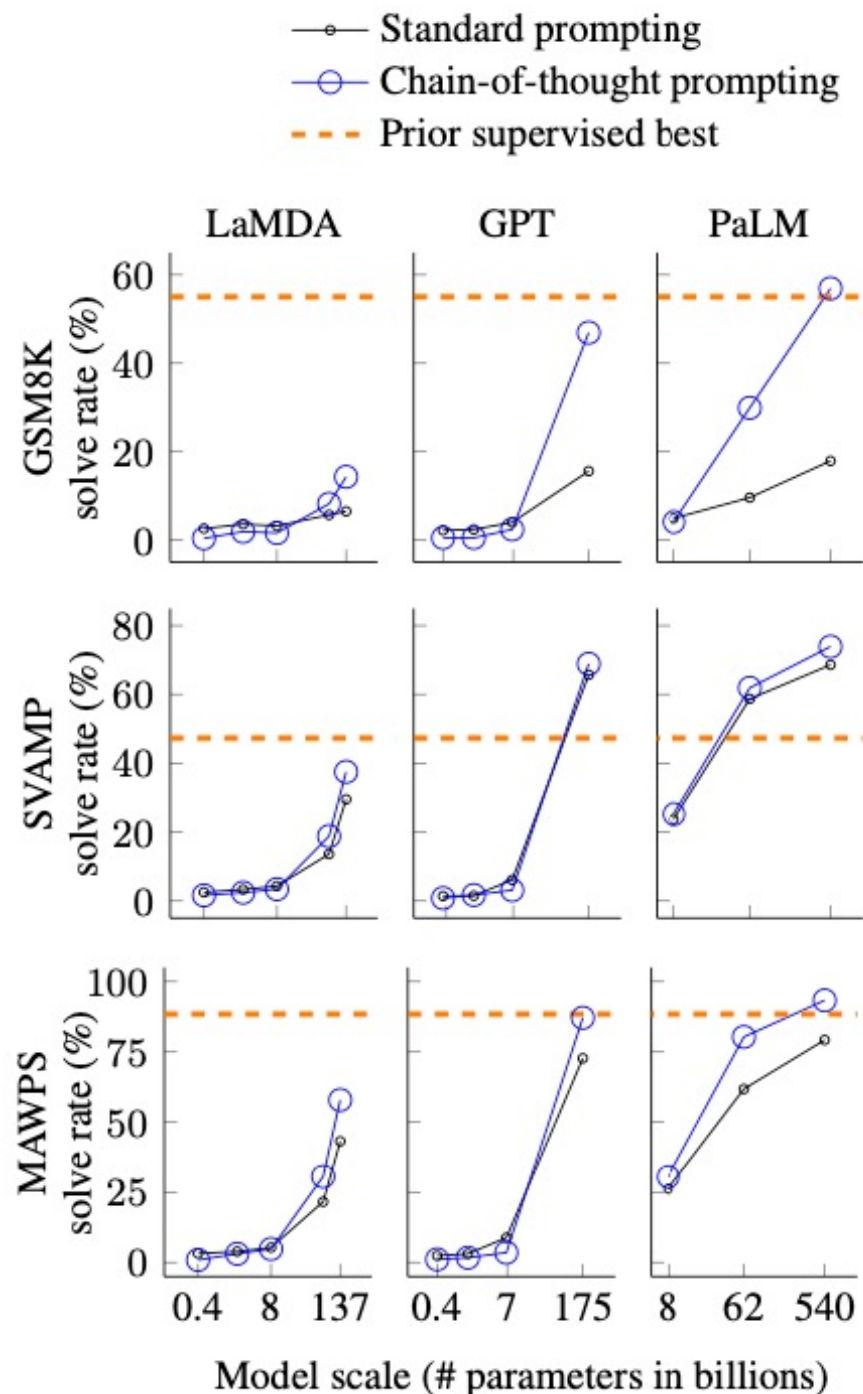


Figure 5: Ablation study for different variations of prompting using LaMDA 137B and PaLM 540B. Results for other datasets are given in Appendix Table 6 and Table 7.



Symbolic Reasoning



LAST LETTER CONCATENATION

Q: Take the last letters of the words in "Elon Musk" and concatenate them.

A: The last letter of "Elon" is "n". The last letter of "Musk" is "k". Concatenating them is "nk". So the answer is nk.

REVERSE LIST

Q: Reverse the sequence "glasses, pen, alarm, license".

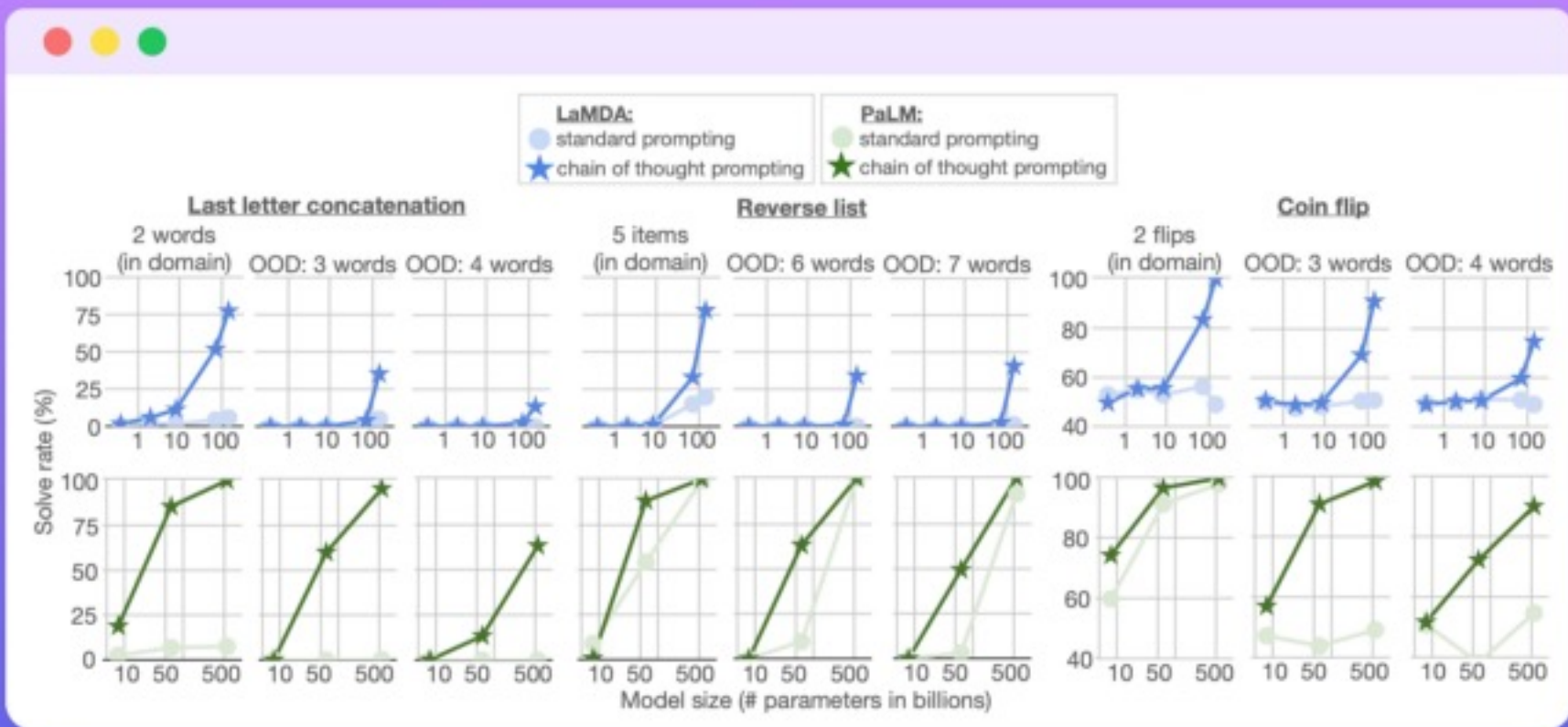
A: First is glasses. Second is pen. Third is alarm. Fourth is license. Now to reverse, change the order to: Fourth is license. Third is alarm. Second is pen. First is glasses. So the answer is "license, alarm, pen, glasses".

COIN FLIP

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

Result



Commonsense Reasoning

- **Benchmarks** : 5 datasets
- **Prompts** : few-shot samplers from 6 ~ 10

COMMONSENSEQA

Q: Sammy wanted to go to where the people were. Where might he go?

Options: (a) race track (b) populated areas (c) desert (d) apartment (e) roadblock

A: The answer must be a place with a lot of people. Race tracks, desert, apartments, and roadblocks don't have a lot of people, but populated areas do. So the answer is (b).

STRATEGYQA

Q: Yes or no: Could a llama birth twice during War in Vietnam (1945-46)?

A: The War in Vietnam was 6 months. The gestation period for a llama is 11 months. So a llama could not give birth twice during the War in Vietnam. So the answer is no.

DATE UNDERSTANDING

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

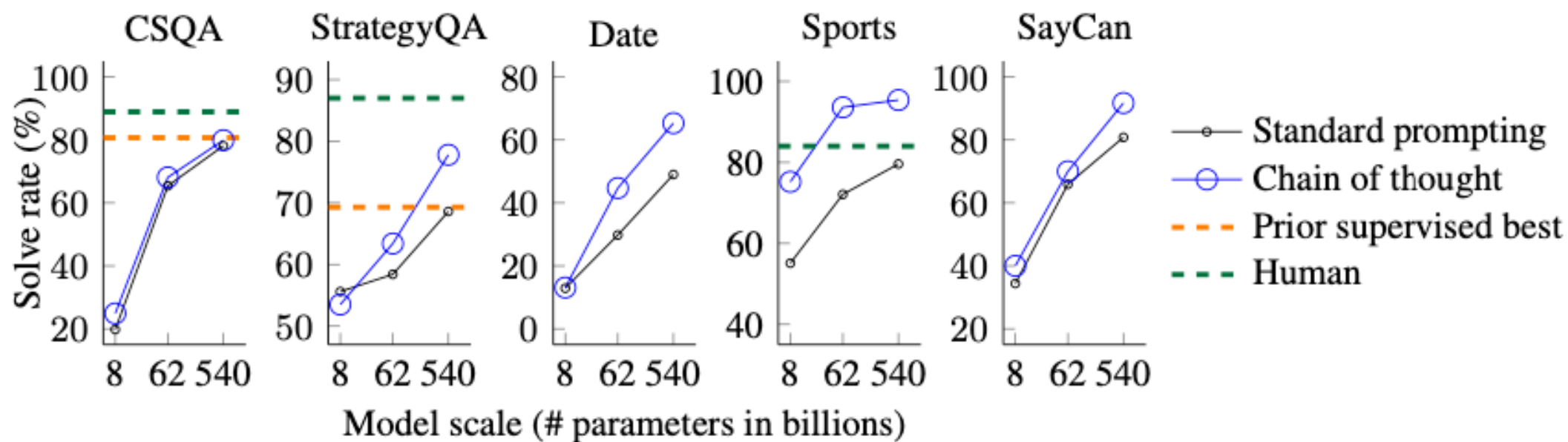
A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.

SPORTS UNDERSTANDING

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.

Result



Discussion

1. Simple mechanism for eliciting multi-step reasoning in large-LM
2. Huge improvement for arithmetic reasoning.
3. Commonsense reasoning is Underscoring how linguistic nature of CoT makes it generally applicable.
4. Facilitates OOD generalization to longer sequence lengths
5. Prompting an off-the-shelf language model.
6. CoT Prompting appears to expand for reasoning successfully and in large LM.

Limitation

- Not answer where the neural network is actually “reasoning”
- Costly augmenting exemplars for CoT minimal in few-shot learning
- No guarantee of correct reasoning paths
- Large LM makes it costly to serve in real-world applications.

References

- **Chain-of-Thought Prompting Elicits Reasoning in Large Language Models**
(<https://arxiv.org/pdf/2201.11903.pdf>)
- **Two minutes NLP – Making Large Language Models reason with Chain of Thought Prompting** (<https://medium.com/nlplanet/two-minutes-nlp-making-large-language-models-reason-with-chain-of-thought-prompting-401fd3c964d0>)

— **THANK YOU**